



2. Strong interaction



STARTSEITE

EINFÜGEN

ZEICHNEN

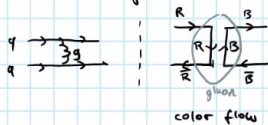
ANSICHT

gluons

- mediator of the strong interaction

$$m_0 = 0$$

- bicolour objects



- colour-anticolour state

3 possibilities

but 1 is colourless

$$|R\bar{R}\rangle + |B\bar{B}\rangle + |G\bar{G}\rangle \Rightarrow \text{gluon does not exist}$$

 $\Rightarrow 8$ gluonsnext time: $SU(2) = \text{isospin}$ $SU(3) = \text{colour, low-mass quarks uds, quark model}$